

Michelle Xu

☎ +1-226-988-5324 | ✉ michelle.xu@uwaterloo.ca | 🔗 linkedin.com/in/xummichelle | 🐙 github.com/xummichelle

Education

University of Waterloo

Bachelor of Computer Science

Expected May 2028

Wilfrid Laurier University

Bachelor of Business Administration

Expected May 2028

Technical Skills

Languages/Frameworks: C++, C#, Python, Golang, Typescript, MySQL

Tools: Unity, Github Actions, Docker, Redis, AWS S3, Git

Work Experience

Bioadaptive Interface Lab

Game Developer and Designer Intern

May 2026 – Aug 2026

- Developed a **research-ready VR rhythm game** within 2 months using **Unity and C#**, ensuring accurate hand tracking, configurable collision detection, and movement-to-music synchrony metrics that enable **data collection** in a 30-participant pilot study.
- Developed a **Python script** to automatically convert MIDI files into rhythm game note-timings for the Unity VR game, streamlining the process of adding rhythm game songs by **75%** for future developers and researchers.
- Built 3 exercise games in 2 months using **C# and Unity**, integrating **MediaPipe computer vision** joint tracking to collect **data** for research in exercise games improving physical mobility for older adults.

RBC

Software Developer Intern

Jan 2025 – Apr 2025

- Developed an **API orchestration** layer using **Typescript** and **Node.js**, enhancing API consumption efficiency and ensuring **data consistency** to prepare for cloud migration.
- Engineered and optimized prompts for Github Copilot to create code comments, documentation, and unit tests, improving developer efficiency in these areas by **90%**.

RBC

SRE Developer Intern

Jan 2024 – Apr 2024

- Developed a Golang service that dynamically updates and returns status images stored in AWS S3, deploying an internal reusable service developers can use for their own status-tracking projects.
- Created Github composite actions that use my service to automatically inject badge images into any repository's README, providing status transparency for each stage of the **Github Actions** CI/CD pipeline used when deploying any RBC application.

RBC

SRE Developer Intern

May 2023 – Aug 2023

- Built a enterprise-wide Webex chatbot used by **200+ employees**, using **Websockets** in **Python**, helping teams access incident information and respond to **incidents 25% faster** and creating a clear foundation for future chatbot developers.
- Developed incident notifications for Webex channels using **Redis** caching and **PostgreSQL** queries, updating teams on major incidents.

RBC

Innovation Developer Intern

Jul 2022 – Aug 2022

- Used **Typescript** and Angular to create a “featured page” of an Innersource navigation platform used daily by development teams, increasing exposure and engagement with repositories by **20%**.
- Organized prototype user testing meetings to identify users' needs and gather UI/UX feedback.

Projects

AI NPC Chatbot Stardew Valley Mod | 🐙 github.com/xummichelle/TalkByMailMod | Azure OpenAI, C#, .NET

July 2026 - Present

- Building an **AI-augmented** NPC dialogue system for a Stardew Valley game extension using .NET and C#, integrating NLP and sentiment analysis with **Azure OpenAI** to generate personality-consistent dialogue and mail responses.

F1 Stats Engine | 🐙 github.com/JSiggelkow/Formula-1-Visualizer | Python, FastAPI, MySQL, React.js, Docker

Sep 2025 - Dec 2025

- Designed and implemented a **MySQL database schema** for F1 data.
- Developed a visualization app that executes complex **MySQL queries** to display race and driver insights.
- Optimized queries using **indexes** after running performance tests.

CC3K Dungeon Crawler Game | C++

July 2024

- Built a roguelike dungeon crawler game along with a design document and UML diagram, focusing on applying **C++ and object-oriented programming** concepts for maximizing extensibility and minimizing code reuse.

Nara Game | 🐙 github.com/nathan8255/Nara | Unity, C#

Nov 2021 – Jan 2022

- Programmed with **Unity** and **C#**, **designed puzzles**, and created pixel art assets for an escape room puzzle game involving movable characters, interactable objects, dialogue, and multiple levels.